# Max Ellis

*BDes (Hons) MSc AFHEA*

## PhD Candidate in Psychoacoustics

*Psychoacoustic Assessment and Machine Learning-Based Modelling of Novel Noise Source Perception in Complex Soundscapes*

ellisacoustics.github.io · linkedin.com/in/psychoacoustic · github.com/ellisacoustics · orcid.org/0009-0007-5909-854X

**Net Zero** technology will bring *sounds* we've never lived with into our towns and cities – **electric cars**, **delivery drones**, **air taxis** and **heat pumps**. I research how these noise sources affect us, *before* they arrive.

Noise is a "silent killer", second only to air pollution in its burden on health in Europe[WHO, 2011, House of Lords, 2023]. The next wave of technology will introduce noise that loudness alone cannot accurately judge: the high-pitched whine of a rotor, the tonal hum of a compressor, rumbles of large aircraft[CAA, 2018, DfT, 2026] flying low over gardens. My PhD research, sponsored by the Department for Environment, Food and Rural Affairs (Defra), seeks to accurately simulate these future soundscapes to gather response data from listening panels, for machine learning models to predict who will be annoyed, to what extent, and why. That gives regulators evidence to set standards on, and engineers a target for quieter machines, before they are everywhere.

## Publications

2027

### Multi-Model Analysis for Prediction of Perceived Annoyance from UAVs

**M. W. Ellis**, T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, Z. Podwińska, A. J. Torija

*Journal of the Acoustical Society of America (JASA), 2027.*

2027

### Psychoacoustic Assessment of UAV Noise in Varying Ambiences, Operational Conditions and Contexts

**M. W. Ellis**, Z. Podwińska, A. J. Torija

*Forum Acusticum 2027, Brussels.*

Awarded Best Student Paper

2027

### Impacts on Soundscape Perception from eVTOL Flyovers in Existing Urban Ambiences

**M. W. Ellis**, N. Shrubsole-Green, J. Bain, Z. Podwińska, A. J. Torija

*AIAA 2027, San Diego.*

2026

### Comparing a General Sound Quality Metric for Aircraft Noise Using a Pre-Trained Neural Network Model to Traditional Psychoacoustics

T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, **M. W. Ellis**, A. J. Torija

*Quiet Drones 2026, Delft. Human Response to UAS and UAM Noise. With HEAD Acoustics GmbH.*

PDF

2025

### Comparison of Deep Learning and Psychoacoustic Models to Predict UAV Noise Impact in Soundscapes

**M. W. Ellis**, M. C. Green, M. J. B. Lotinga, A. J. Torija

*Forum Acusticum EuroNoise 2025, 11th Convention of the European Acoustics Association, Málaga, pp. 2121–2127.*

doi:10.61782/fa.2025.0235 · PDF

All publications on Google Scholar · ORCID

## Industry Experience

Aug 2027 – Oct 2027

### Research Engineer (PhD Placement)

Dyson

- Assessing whether structural vibration can underpin sound quality metrics for product components, comparing vibration and sound quality data across anechoic, semi-reverberant and reverberant conditions.
- Applying transfer learning across sound quality metrics and jury test data from other products, to build machine learning models that predict listener response and satisfaction.

Aug 2026 – Oct 2026

### Specialist Consultant (PhD Placement)

Noise Consultants Ltd. (part of Logika Group)

- Validated open-source noise measurements against modelled Defra noise-mapping data for infrastructure assessment.
- Ran a field campaign on an airport expansion project comparing physical and perceptual dominance of aircraft noise.

## Academic Experience

Sep 2026 – May 2027

### Visiting Lecturer in Music Technology

University of the West of Scotland

- 1st Supervisor on the Major Project (Dissertation) module for BSc (Hons) Music Technology students specialising in Acoustics and Environmental Noise.

2026 – Present

### Associate Lecturer in Computer Science

St Mary’s University, Twickenham

- Teaching, marking and moderation across the online MSc Computer Science suite.
- Module reviewer and author.

2026 – Present

### Teaching Assistant in Applied Acoustics

University of Salford

- Lab demonstrator for Environmental Noise Measurement and Modelling (BEng/MSc).
- Tutor for Immersive Sound Reproduction (MSc/PgDip).

## Education

2025 – 2028

### PhD Acoustics and Audio Engineering

University of Salford Acoustics Innovation Institute

**Year 2**

2023 – 2024

### MSc Computer Science with Artificial Intelligence

Abertay University

**Distinction**

2021 – 2023

### BDes (Hons) Sound for Moving Image

University of Glasgow

**2nd Class Honours, Upper-Division (2:1)**

2019 – 2021

### DipHE Music Technology

University of the West of Scotland

**Distinction**

🎖 Awarded BSc Level 8 Court Medal

2019

### BTEC Level 3 Digital Synthesis and Sampling

Fife College

**Pass**

2018 – 2019

### SCQF Level 6 Sound Production

Glasgow Clyde College Anniesland

**Pass**

## Working Groups and Committees

**Institute of Acoustics** Scottish Branch Committee — Member; Early Careers Group — Member

**NASA** Unmanned Aircraft Systems Noise Working Group (UNWG) — Member

**Noise Network Plus** A.I. and Digital Working Group — Member

## Software Proficiencies

Max/MSP Ableton ArtemiS SUITE R Python QGIS LaTeX (Overleaf) REAPER MATLAB TensorFlow PyTorch Feature Manipulation Engine (FME)

## Certifications

**Advanced Prompt Engineering**
Vanderbilt University
Show credential ↗

**Advanced R Programming**
Johns Hopkins University

**AI for Noise Pollution Control**
LSIB
Credential ID 2245-32007-113

**Applications of AI in Acoustics**
European Acoustics Association Summer School

**Auralization**
Autumn School Series in Acoustics (ASSA) 2026

**Digital Signal Processing**
EPFL
Show credential ↗

**Ethics of Artificial Intelligence**
Politecnico di Milano
Show credential ↗

**Responsible Research and Innovation**
ORBITRRI

**Transformer Models and BERT Model**
Google

**Trustworthy Generative AI**
Vanderbilt University
Show credential ↗

## Research Collaborations

**Defra** — PhD Sponsor
**HEAD Acoustics GmbH** — Journal Paper
**Joby Aviation Ltd.** — Journal Paper